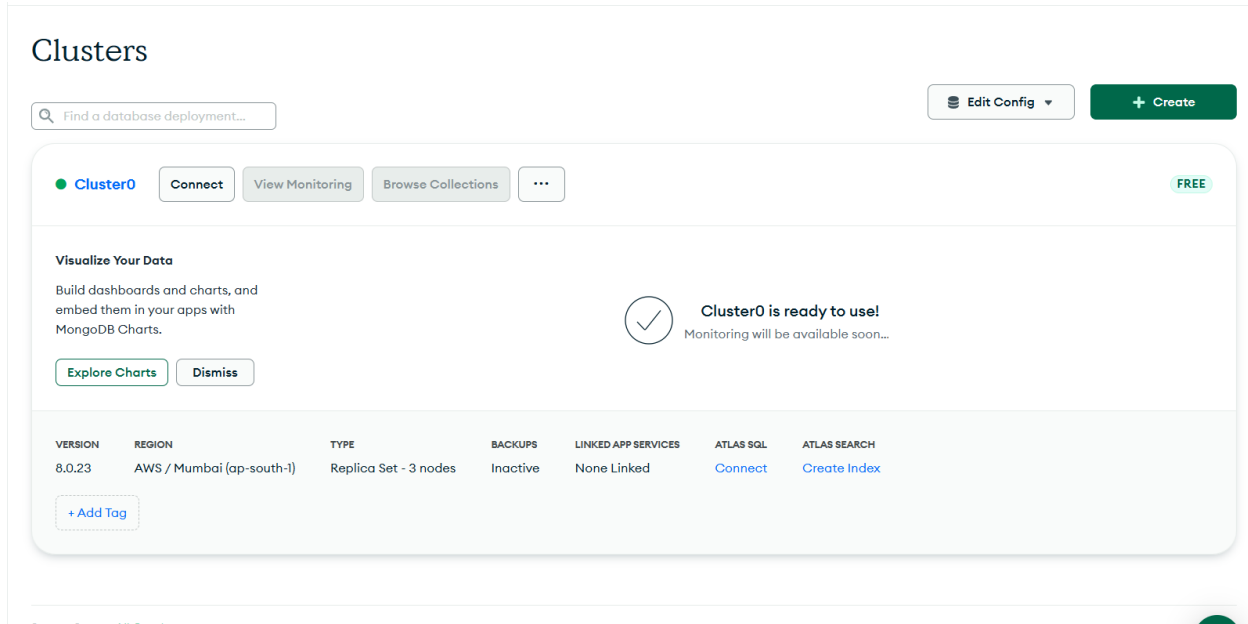


2. Creating the cluster



The screenshot shows the MongoDB Atlas 'Clusters' management interface. At the top, there's a search bar and buttons for 'Edit Config' and '+ Create'. Below this, a card for 'Cluster0' is displayed, featuring buttons for 'Connect', 'View Monitoring', 'Browse Collections', and a 'FREE' label. A message states 'Cluster0 is ready to use!' with a checkmark icon and a note that 'Monitoring will be available soon...'. Below the message is an 'Explore Charts' button. A table lists cluster details:

VERSION	REGION	TYPE	BACKUPS	LINKED APP SERVICES	ATLAS SQL	ATLAS SEARCH
8.0.23	AWS / Mumbai (ap-south-1)	Replica Set - 3 nodes	Inactive	None Linked	Connect	Create Index

A '+ Add Tag' button is located below the table.

Connected to mongoshell

```
C:\Users\ADMIN>mongosh "mongodb+srv://cluster0.ypvynfo.mongodb.net/" --apiVersion 1 --username admin
Enter password: *****
Current Mongosh Log ID: 69fec90341f9c8f0097c2906
Connecting to:      mongodb+srv://<credentials>@cluster0.ypvynfo.mongodb.net/?appName=mongosh+2.7.0
Using MongoDB:      8.0.23 (API Version 1)
Using Mongosh:       2.7.0
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-wi47rr-shard-0 [primary] test>
```

Insert the data

```
Atlas atlas-wi47rr-shard-0 [primary] test> use ecommerce
switched to db ecommerce
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.createCollection(orders)
ReferenceError: orders is not defined
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.createCollection('orders')
{ ok: 1 }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> show collections
orders
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.insertMany([
|   { "order_id": "ORD001", "customer": "Alice", "city": "Mumbai",
|     "category": "Electronics", "product": "Laptop", "quantity": 1,
|     "price": 75000, "status": "delivered", "date": "2024-01-15" },
|   { "order_id": "ORD002", "customer": "Bob", "city": "Delhi",
|     "category": "Clothing", "product": "Jacket", "quantity": 3,
|     "price": 4500, "status": "pending", "date": "2024-02-20" },
|   { "order_id": "ORD003", "customer": "Carol", "city": "Mumbai",
|     "category": "Electronics", "product": "Phone", "quantity": 2,
|     "price": 25000, "status": "delivered", "date": "2024-03-10" },
|   { "order_id": "ORD004", "customer": "Dave", "city": "Pune",
|     "category": "Books", "product": "Big Data Guide", "quantity": 5,
|     "price": 800, "status": "shipped", "date": "2024-03-25" },
|   { "order_id": "ORD005", "customer": "Eve", "city": "Delhi",
|     "category": "Electronics", "product": "Tablet", "quantity": 1,
|     "price": 35000, "status": "delivered", "date": "2024-04-05" }
| ])
| {
|   acknowledged: true,
|   insertedIds: {
|     '0': ObjectId('69fec9e141f9c8f0097c2907'),
|     '1': ObjectId('69fec9e141f9c8f0097c2908'),
|     '2': ObjectId('69fec9e141f9c8f0097c2909'),
|     '3': ObjectId('69fec9e141f9c8f0097c290a'),
|     '4': ObjectId('69fec9e141f9c8f0097c290b')
|   }
| }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

CRUD operations :

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.find({status:"delivered"},{_id:0})
[
  {
    order_id: 'ORD001',
    customer: 'Alice',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Laptop',
    quantity: 1,
    price: 75000,
    status: 'delivered',
    date: '2024-01-15'
  },
  {
    order_id: 'ORD003',
    customer: 'Carol',
    city: 'Mumbai',
    category: 'Electronics',
    product: 'Phone',
    quantity: 2,
    price: 25000,
    status: 'delivered',
    date: '2024-03-10'
  },
  {
    order_id: 'ORD005',
    customer: 'Eve',
    city: 'Delhi',
    category: 'Electronics',
    product: 'Tablet',
    quantity: 1,
    price: 35000,
    status: 'delivered',
    date: '2024-04-05'
  }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.updateOne({order_id:'ORD002'},{$set:{status:'shipped'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.insertOne({ "order_id": "ORD006", "customer": "Jay", "city": "Mumbai",
|   "category": "Electronics", "product": "Laptop", "quantity": 1,
|   "price": 90000, "status": "shipped", "date": "2024-05-06" })
{
  acknowledged: true,
  insertedId: ObjectId('69fecc4e41f9c8f0097c290c')
}
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.deleteOne({order_id:'ORD004'})
{ acknowledged: true, deletedCount: 0 }
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

Aggregate pipeline:

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group:{_id:"$category", total_revenue:{$sum:{$multiply:[$price,$quantity]} }},{$sort:{total_revenue:-1}}}]
[
  { _id: 'Electronics', total_revenue: 250000 },
  { _id: 'Clothing', total_revenue: 13500 },
  { _id: 'Books', total_revenue: 4000 }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group:{_id:'$city', total_orders:{$sum:1}, avg_price:{$avg:'$price'}}}]
[
  { _id: 'Mumbai', total_orders: 3, avg_price: 63333.333333333336 },
  { _id: 'Delhi', total_orders: 2, avg_price: 19750 },
  { _id: 'Pune', total_orders: 1, avg_price: 800 }
]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

```
Atlas atlas-wi47rr-shard-0 [primary] ecommerce> db.orders.aggregate([{$group:{_id:'$status', count:{$sum:1}}, {$match:{count:{$gt:1}}}]
[ { _id: 'delivered', count: 3 }, { _id: 'shipped', count: 3 } ]
Atlas atlas-wi47rr-shard-0 [primary] ecommerce>
```

1. Create an EMR cluster

Amazon EMR > EMR on EC2: Clusters > my_practiacl_cluster_43

Your cluster 'my_practiacl_cluster_43' has been successfully created.

my_practiacl_cluster_43 Updated 7 minutes ago [Terminate](#) [Clone in AWS CLI](#) [Clone](#)

This EMR release reaches End of Support on Aug-01-2026 and will no longer be eligible for technical support. AWS strongly recommends that you run your workloads on the latest Amazon EMR release to receive security-critical updates and fixes. You can also use new Spark upgrade agent to modernize your existing applications on version 5.40 or higher to latest EMR version. To learn more, see [EMR Standard Support policy](#) and [Spark Upgrades](#).

▼ Summary

Cluster info

Cluster ID
j-2OUO6P68CH4RN

Cluster ARN
arn:aws:elasticmapreduce:us-east-1:4711127535:24:cluster/j-2OUO6P68CH4RN

Cluster configuration
Instance groups

Capacity
1 Primary | 1 Core | 0 Task

Applications

Amazon EMR version
emr-5.20.1

Installed applications
Ganglia 3.7.2, Hadoop 2.8.5, Hive 2.3.4, Hue 4.3.0, Tez 0.9.1

Cluster management

Log destination in Amazon S3
Logging not configured

Primary node public DNS
ec2-100-27-2-158.compute-1.amazonaws.com
[Connect to the Primary node using SSH](#)
[Connect to the Primary node using SSM](#)

Status and time

Status
Waiting

Creation time
9 May 2026 11:02 (UTC+05:30)

Elapsed time
48 minutes, 29 seconds

Properties

Bootstrap actions

Instances (hardware)

Steps

Applications

Configurations

Monitoring

Events

Tags (0)

Connected to EMR cluster via MobaXterm

```

• MobaXterm Personal Edition v26.3 •
(SSH client, X server and network tools)

▶ SSH session to hadoop@ec2-100-27-2-158.compute-1.amazonaws.com
  • Direct SSH      : ✓
  • SSH compression : ✓
  • SSH-browser     : ✓
  • X11-forwarding  : ✗ (disabled or not supported by server)

▶ For more info, ctrl+click on help or visit our website.

```

Last login: Sat May 9 06:24:15 2026

```

 _ | _ | )
 _ | ( _ /   Amazon Linux AMI

```

https://aws.amazon.com/amazon-linux-ami/2018_03-release-notes/

54 package(s) needed for security, out of 80 available
Run "sudo yum update" to apply all updates.

```

EEEEEEEEEEEEEEEEEE MMMMMMMM                MMMMMMMM RRRRRRRRRRRRRR
E:::EEEEEEEEEE:::E M:::MM          M:::MM R:::R:::R:::R
E:::EEEEEEEEEE:::E M:::MM          M:::MM R:::RARRRR:::R
 E:::E         EEEEE M:::MM        M:::MM RR:::R    R:::R
E:::E           M:::MM        M:::MM RR:::R    R:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM R:::RARRRR:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM R:::R:::R:::R
E:::EEEEEEEEEE M:::MM M:::MM M:::MM R:::RARRRR:::R
E:::E           M:::MM M:::MM M:::MM R:::R    R:::R
E:::E         EEEEE M:::MM   MMM   M:::MM R:::R    R:::R
EE:::EEEEEEEEEE:::E M:::MM          M:::MM R:::R    R:::R
E:::EEEEEEEEEE:::E M:::MM          M:::MM RR:::R    R:::R
EEEEEEEEEEEEEEEEEE MMMMMMMM                MMMMMMMM RRRRRRR    RRRRRR

```

[hadoop@ip-172-31-63-182 ~]\$ █

Connected to hue

The screenshot shows the Apache Hue web interface. At the top, there's a navigation bar with the Hue logo, a 'Query' dropdown menu, a search bar labeled 'Search saved documents...', and links for 'Jobs', a refresh icon, and a user profile labeled 'hadoop'. Below this is the 'File Browser' section. On the left sidebar, it says '< default', 'Tables', '(0) + ↻', and 'No entries found'. The main area has a breadcrumb path 'Home / user / hadoop'. There are two buttons at the top right: 'Upload' and 'New'. A table lists files and directories:

☐	Name	Size	User	Group	Permissions	Date
☒	j		hdfs	hadoop	drwxr-xr-x	May 08, 2026 10:37 PM
☐	.		hadoop	hadoop	drwxrwxrwx	May 08, 2026 10:37 PM

At the bottom, it says 'Show 45 of 0 items' and 'Page 1 of 1' with pagination controls.

Created external table and loaded csv file into table:

```
hive> CREATE EXTERNAL TABLE crimes_raw (
>   crime_id BIGINT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   iucr STRING,
>   primary_type STRING,
>   description STRING,
>   location_description STRING,
>   arrest BOOLEAN,
>   domestic BOOLEAN,
>   beat INT,
>   district INT,
>   ward INT,
>   community_area INT,
>   fbi_code STRING,
>   x_coordinate BIGINT,
>   y_coordinate BIGINT,
>   year INT,
>   updated_on STRING,
>   latitude DOUBLE,
>   longitude DOUBLE,
>   location STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
OK
Time taken: 0.831 seconds
hive> load data inpath '/user/hadoop/Chicago_Crimes.csv' into table crimes_raw;
Loading data to table crime_db.crimes_raw
OK
Time taken: 1.418 seconds
hive> 
```

3.

The screenshot shows the Databricks Catalog Explorer interface. The left sidebar contains navigation options: Home, Learn, Workspace, Recents, Catalog (selected), Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, and Query History. The main panel displays the 'my_practicals_43' catalog, which is a Serverless Starter Warehouse. The 'Catalog Explorer' shows a tree view of the catalog's contents, including 'workspace', 'system', 'my_practicals_43' (selected), 'default', 'information_schema', 'spark_assignment', 'vita', 'Delta Shares Received', and 'samples'. The 'Overview' tab is active, showing a description of the catalog and a table of schemas. The table has columns for Name, Owner, and Created at. The schemas listed are 'default' (owned by dandekarviraj1999@gmail.co...) and 'information_schema' (owned by System user), both created on May 09, 2026, at 12:05 PM. The 'About this catalog' section shows the owner as dandekarviraj1999@gmail.com. The 'Tags' section has an 'Add tags' button.

Name	Owner	Created at
default	dandekarviraj1999@gmail.co...	May 09, 2026, 12:05 PM
information_schema	System user	May 09, 2026, 12:05 PM

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

Catalog

Serverless Starter Warehouse Serverless 2XS

Type to search...

For you All

My organization

- workspace
- system
- my_practicals_43
 - bigdata_practicals
 - bigdata

Catalog Explorer > my_practicals_43 > bigdata_practicals >

bigdata

Overview Files Details Permissions

Description

AI generate Add

/Volumes/my_practicals_43/bigdata_practicals/bigdata Refresh Create directory

Filter files and directories at this level

Name	Size	Last modified
No content in volume		

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Catalog

Serverless Starter Warehouse Serverless 2XS

Type to search...

For you All

My organization

- workspace
- system
- my_practicals_43
 - bigdata_practicals

Catalog Explorer > my_practicals_43 >

bigdata_practicals

Overview Details Permissions Policies

Description

AI generate Add

Filter tables Tables 0 Volumes 0 Models 0 Functions 0

Sort

Name	Owner	Created at	Popularity	Comments
You either don't have access to any tables or there are no tables in this schema. Either contact administrator or create a new sample table.				

Catalog Explorer > my_practicals_43 > bigdata_practicals >

 **bigdata**   ⋮ Share Upload to this volume

Overview Files Details Permissions

Description

 AI generate Add

/Volumes/my_practicals_43/bigdata_practicals/bigdata  Refresh Create directory

 Filter files and directories at this level

Name	Size	Last modified
 airports.csv	12.03 MB	13 seconds ago
 runways.csv	3.76 MB	18 seconds ago

About this volume

Owner [dandekarviraj1999@gmail.com](#) 

26044525043_Viraj_Dandekar × +

File Edit View Run Help Python ☆ Last edit was now Run all Serverless Schedule Share

2 minutes ago (<1s) 1

```
import pyspark
```

Just now (3s) 2 Python 🗑️ 🔗 🔍 ⋮

```
df_airports = spark.read.csv('/Volumes/my_practicals_43/bigdata_practicals/bigdata/airports.csv', header=True, inferSchema=True)
df_runways = spark.read.csv('/Volumes/my_practicals_43/bigdata_practicals/bigdata/runways.csv', header=True, inferSchema=True)
```

>  df_airports: pyspark.sql.connect.dataframe.DataFrame = [id: integer, ident: string ... 17 more fields]

>  df_runways: pyspark.sql.connect.dataframe.DataFrame = [id: integer, airport_ref: integer ... 18 more fields]

